

Graph Neural Networks for Context-Aware Recommendations under Context-Sensitive Evaluation Dimensions

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Abstract

Context-aware recommender systems aim to improve personalization by considering situational factors such as time or location, but effectively modeling this information and evaluating its impact remain challenging. At the same time, Graph Neural Networks (GNNs) have shown strong performance in capturing complex relationships in recommendation tasks, although their use in context-aware scenarios is still limited. Motivated by this research gap, we propose a novel GNN-based model that integrates contextual information into graph structures and message-passing mechanisms. In addition, we introduce novel offline evaluation metrics that capture context-sensitive aspects such as relevance, coverage, and diversity. Experimental results on three benchmark datasets show that our model outperforms existing baselines in terms of both accuracy and context-aware metrics. The complete code is publicly available at <https://anonymous.4open.science/r/Graph-CARS-9DDE>

CCS Concepts

• Information systems → Recommender systems; Evaluation of retrieval results; • Human-centered computing → Contextual design.

Keywords

Context-aware recommender systems, GNN, evaluation

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1 Introduction

Context-aware recommender systems (CARS) constitute a well-established research area that has been extensively studied for more than two decades [1]. Their core objective is to enhance the relevance and personalization of recommendations by exploiting situational (contextual) information—such as time, location, or social settings. Unlike traditional recommendation approaches that primarily rely on user-item interactions (e.g., ratings), CARS model the dynamic nature of user preferences, acknowledging that user interests may

vary depending on the surrounding circumstances [2]. This capability provides important benefits, including greater adaptability to real-world scenarios and increased user satisfaction. Nevertheless, CARS also pose significant research challenges, such as the representation of heterogeneous contextual signals, the sparsity of context data, and the complexity of modeling and exploiting high-dimensional interactions [3]. Moreover, evaluating context-aware recommendations remains an open problem, since conventional accuracy-oriented metrics often fail to capture the actual contextual relevance of recommendations [4].

In parallel, graph-based recommender systems—particularly those based on Graph Neural Networks (GNNs)—have demonstrated strong capabilities for modeling complex relational structures [5, 6]. By representing users, items, and their interactions as graph entities, GNNs can capture high-order connectivity patterns and learn expressive latent representations that significantly improve recommendation performance. Consequently, GNN-based recommendation approaches have attracted considerable attention and achieved state-of-the-art results across multiple application domains [7].

Despite these advances, the integration of contextual information into GNN-based recommender systems remains relatively underexplored [3]. Existing approaches still face important limitations regarding both the incorporation of contextual signals into graph learning processes and the evaluation of recommendations under different contextual conditions. In particular, most evaluation frameworks continue to rely mainly on traditional accuracy metrics, overlooking context-sensitive aspects such as the coverage of target contextual situations and the contextual diversity of recommendations [7].

To address these challenges, this paper proposes a novel context-aware recommendation model based on GNNs that explicitly incorporates contextual information into the graph learning process. Furthermore, we introduce a set of context-sensitive evaluation metrics designed to better assess recommendation effectiveness under varying contextual conditions. Finally, we conduct an extensive experimental evaluation comparing the proposed approach against state-of-the-art baseline recommenders on three benchmark datasets: Frappe, a widely used dataset for context-aware mobile application recommendations; Yelp, a standard benchmark for location-based business recommendations; and BoardGameGeek, a dataset focused on board game recommendations under different playing scenarios. Experimental results show that our model improves recommendation quality not only in terms of traditional accuracy metrics, but also regarding context-aware evaluation dimensions.

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2 Related Work

In this section, ...

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2.1 Context-Aware Recommender Systems

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The literature identifies three principal paradigms for considering context in the recommendation process [2, 8]: *contextual pre-filtering*, *contextual post-filtering*, and *contextual modeling*. *Contextual pre-filtering* exploits contextual information to select or construct the subset of interactions that are most relevant to the current situation before applying a conventional recommendation algorithm. In contrast, *contextual post-filtering* first generates recommendations independently of the contextual conditions and subsequently re-ranks or filters the resulting recommendation list according to contextual signals. Finally, *contextual modeling* incorporates contextual variables into the prediction function itself, enabling the learning of context-dependent representations in an end-to-end manner.

These paradigms exhibit different trade-offs in terms of flexibility, expressiveness, scalability, and computational complexity. Pre-filtering methods are conceptually simple and allow the reuse of traditional recommendation algorithms; however, partitioning interactions according to contextual conditions often aggravates data sparsity problems. Post-filtering techniques alleviate this issue to some extent, since they operate on recommendations generated from the complete interaction space, but contextual information only affects the final ranking stage and is therefore not fully exploited during representation learning. By contrast, contextual modeling approaches integrate context into the learning mechanism itself, allowing the system to capture complex interactions among users, items, and contextual dimensions. Although these approaches generally require more sophisticated architectures and greater computational resources, they have demonstrated superior capability in modeling dynamic user preferences and producing higher-quality recommendations [4, 8]. For the contextual modeling paradigm, Factorization Machines (FM) [9] and their extensions [10–13] represent one of the most widely adopted models for CARS. They naturally incorporate heterogeneous contextual features as additional input fields alongside user and item identifiers, enabling the modeling of pairwise and higher-order feature interactions while maintaining scalability on sparse recommendation datasets. As a result, FM have established strong performance baselines across context-aware recommendation tasks.

The application of GNNs to contextual modeling remains comparatively limited. Most existing GNN-based recommender systems focus primarily on modeling user–item interaction graphs, while contextual information is either ignored or incorporated only as auxiliary node attributes [5, 7]. Consequently, many current approaches fail to fully exploit the relational dependencies among users, items, and contextual conditions, particularly in scenarios involving heterogeneous or multi-dimensional contexts.

In this paper, we address research gap by proposing a context-aware recommendation model based on GNNs that explicitly incorporates contextual entities and relations into the graph representation learning process. In contrast to approaches that primarily treat context as side information, our model integrates contextual conditions directly into the graph structure, enabling the learning of richer and more expressive context-dependent representations.

2.2 GNN-Based Recommender Systems

Graph neural networks have emerged as a dominant paradigm in collaborative filtering due to their ability to capture high-order connectivities within user–item bipartite graphs by propagating embeddings on them [14].

Early GNN-based models inherited operations originally designed for attributed graphs, such as feature transformations and non-linear activations, which proved to be burdensome and largely unnecessary for recommendation tasks where nodes are represented solely by ID embeddings. To address this, LightGCN [15] proposed a drastically simplified architecture that retains only the most essential component of GCNs: neighborhood aggregation. By linearly propagating embeddings and combining them across layers via a weighted sum, LightGCN achieves superior recommendation performance and training efficiency while effectively avoiding the complexity of standard graph convolutions.

More recently, Graph Contrastive Learning (GCL) has been adopted to alleviate data sparsity by extracting self-supervised signals, typically through data augmentation techniques applied alongside a GNN encoder. Challenging this standard paradigm, LightCCF [16] theoretically demonstrated that the gradient descent process of the standard InfoNCE loss is mathematically equivalent to a step of graph convolution. Building upon this insight, LightCCF abandons both data augmentation and the GCN encoder entirely, introducing a novel Neighborhood Aggregation (NA) loss that treats observed user–item interactions as positive pairs. This objective pulls users closer to their interacted items while pushing them away from other positive pairs within the batch, achieving high-quality neighborhood aggregation while preventing the over-smoothing issues associated with repeated message-passing layers.

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2.3 Evaluation of Recommender Systems

Iván: I would remove this subsection, and check whether its references should be cited in other sections of the paper. Alejandro: I agree. This section is not needed to understand the proposed metrics. This motivation can be made in the introduction instead. If other context-sensitive metrics exist, then this section should list and describe them, but since this is not the case (beyond satisfaction) this section probably makes little sense

Offline evaluation of recommender systems is typically done by ranking, i.e. given a set of test queries, a model produces a ranked list of items, and quality is assessed by comparing the list against held-out ground-truth interactions [17, 18]. The most common metrics used are nDCG@K, MAP@K, Recall@K, and MRR@K, which operate at the level of the ranked list, rewarding systems that surface relevant items at high positions [17].

These measures all have one key assumption in common: a user’s relevance for an item is only a function of the user–item pair and does not take the situational context in which the recommendation is made into account. Therefore, in the case of CARS, a model that recommends the right item but the wrong context is basically treated as a model that recommends the same item in the right context when it comes to evaluation [2].

This limitation motivates the context-sensitive evaluation metrics introduced in Sec. 4.

3 LightCCF-CA: A Context-Aware GNN Recommendation Model

In this section, we introduce LightCCF-CA, a context-aware recommendation model derived from LightCCF [16]. We begin by formulating the problem as a context-enriched bipartite graph, where each user-item interaction is associated with a contextual feature vector (Section 3.1). Building on this formulation, we present the contextual edge embedding module, which projects interaction contexts into the shared node embedding space (Section 3.2). We then describe the context-gated message passing mechanism, in which contextual edge embeddings act as feature-wise filters during neighborhood aggregation, followed by the layer combination strategy (Sections 3.3–3.4). Finally, we define the training objective of LightCCF-CA in Section 3.5. I would simplify the structure of the section to 3 subsections: Contextual Graph Representation (3.1 + 3.2), Context-Gated Message Passing (3.3 + 3.4), Context-Aware Prediction and Training Objective (3.5)

3.1 Problem Formulation

Let $\mathcal{U}$ and $\mathcal{V}$ denote the sets of users and items, respectively. User-item interactions are modeled as a bipartite graph $\mathcal{G} = (\mathcal{U} \cup \mathcal{V}, \mathcal{E})$, where each edge $e_{uv} \in \mathcal{E}$ connects a user $u \in \mathcal{U}$ to an item $v \in \mathcal{V}$, and is annotated with a contextual feature vector obtained from contextual attributes which characterize the interaction (e.g., city, weekday, cost). Unlike approaches that flatten context into node features or scalar edge weights, our model treats each edge as a first-class relational object carrying rich situational information.

3.2 Contextual Edge Embedding

Each contextual feature $f \in \{1, \dots, F\}$ in the interaction between user $\mathcal{U}$ and item $\mathcal{V}$ takes a discrete value $c_{uv}^{(f)}$ drawn from a vocabulary of size V_f (e.g., weekday takes a value from the set $\{0, \dots, 6\}$ of size 7). We assign a dedicated embedding matrix $E_f \in \mathbb{R}^{V_f \times d_c}$ to each feature, mapping its value to a dense vector of size d_c :

$$\mathbf{e}_{uv}^{(f)} = E_f[c_{uv}^{(f)}] \in \mathbb{R}^{d_c}. \quad (1)$$

The F feature embeddings are concatenated and projected into the shared node embedding space of dimension d through a linear layer followed by a ReLU activation:

$$\mathbf{r}_{uv} = \text{ReLU}(\mathbf{W}_c [\mathbf{e}_{uv}^{(1)} \parallel \dots \parallel \mathbf{e}_{uv}^{(F)}] + \mathbf{b}_c) \in \mathbb{R}^d, \quad (2)$$

where $\mathbf{W}_c \in \mathbb{R}^{d \times F \times d_c}$ and $\mathbf{b}_c \in \mathbb{R}^d$ are learnable parameters. The resulting vector $\mathbf{r}_{uv}$ serves as the contextual edge representation for the interaction (u, v) .

3.3 Context-Gated Message Passing

Standard LightGCN [15] propagates plain node embeddings along edges without any relational modulation. Inspired by the concept of Feature Preference Aware Message passing (FPAMP) introduced in EdgeGFL [19], we augment messages propagation with context-gated messages: rather than passing the neighbor's unmodified representation, the contextual edge vector $\mathbf{r}_{uv}$ acts as a *multi-channel multiplicative filter* that selectively scales each dimension of the propagated embedding. Formally, the message from item v to user u at layer l is:

$$\mathbf{M}_{v \rightarrow u}^{(l)} = \mathbf{h}_v^{(l)} \odot \mathbf{r}_{uv}, \quad (3)$$

where $\odot$ denotes the Hadamard (element-wise) product. The same formulation applies symmetrically for messages from users to items. Each dimension of the neighbor's representation is independently scaled by the corresponding dimension of the contextual edge vector, allowing the model to weight information propagation according to the situational context of each recorded interaction. Node representations are updated by summing all incoming context-gated messages from the neighborhood $\mathcal{N}_u$, according to the aggregation proposed in EdgeGFL:

$$\mathbf{h}_u^{(l+1)} = \sum_{v \in \mathcal{N}_u} \mathbf{h}_v^{(l)} \odot \mathbf{r}_{uv}. \quad (4)$$

Following the LightGCN convention, no non-linear activation or feature transformation is applied during aggregation, preserving linearity and scalability.

3.4 Layer Combination and Final Representations

Let L be the total number of propagation layers. After the L -th layer, the representations from all layers (including the initial embeddings) are combined via mean pooling to obtain the final user and item representations:

$$\mathbf{h}_u^* = \frac{1}{L+1} \sum_{l=0}^L \mathbf{h}_u^{(l)}, \quad \mathbf{h}_v^* = \frac{1}{L+1} \sum_{l=0}^L \mathbf{h}_v^{(l)}. \quad (5)$$

This multilayer combination captures both local (low-order) and global (high-order) collaborative signals, a design choice well established in the GNN-based recommendation literature [15]. Furthermore, if we increase the number of layers, the embeddings would be over-smoothed [20], so it is preferable not to use the representations returned by the last layer of the GNN.

3.5 Context-Aware Scoring and Training

Scoring. At inference time, the relevance score between user u and item v under a query context $\mathbf{c}_q$ is computed as:

$$\hat{y}_{uv} = \mathbf{h}_u^* \cdot \mathbf{h}_v^* + \mathbf{r}_q \cdot \mathbf{h}_v^*, \quad (6)$$

where $\mathbf{r}_q = \text{ReLU}(\mathbf{W}_c [\mathbf{e}_q^{(1)} \parallel \dots \parallel \mathbf{e}_q^{(F)}] + \mathbf{b}_c)$ is the embedding of the query context computed via the same encoder defined in Eq. (2). The first term captures the general user-item affinity learned through context modulated graph propagation; the second provides an explicit context-item compatibility signal that adjusts scores at prediction time according to the situational conditions under which the recommendation is requested.

Training Objective. We adopt the multi-task joint training objective introduced by LightCCF [16], which combines three complementary loss terms:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \alpha \mathcal{L}_{\text{NA}} + \lambda \|\mathbf{E}_0\|_F^2, \quad (7)$$

where α and λ are tunable hyperparameters, and $\mathbf{E}_0$ denotes the base embedding parameters subject to L2 regularization.

The **BPR loss** [21] serves as the primary loss to guide the optimization of pairwise preferences over the context-aware scores of Eq. (6):

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u, v^+, v^-) \in \mathcal{D}} \log \sigma(\hat{y}_{uv^+} - \hat{y}_{uv^-}), \quad (8)$$

with $\mathcal{D}$ denoting the set of training triples composed of a user u , a positive item v^+ (observed interaction), and a sampled negative item v^- . Note that we use the same context to compute the score for both positive and negative item, i.e., the context of the positive interaction. The **Neighborhood Aggregation (NA) loss** complements BPR by directly optimizing the mutual information between user and all of their interacted items, without relying on graph convolution for neighborhood aggregation. Given a user u with interacted item set $\mathcal{N}_u$, the positive pairs are defined as user–item pairs (u, v^+) with $v^+ \in \mathcal{N}_u$, while the negative pairs are the remaining user–item pairs in the batch $\mathcal{B}$, excluding those involving u itself:

$$\mathcal{L}_{\text{NA}} = - \sum_{(u, v^+) \in \mathcal{B}} \log \frac{\exp(\text{sim}(\mathbf{h}_u^*, \mathbf{h}_{v^+}^*)/\tau)}{\sum_{p \in \mathcal{B}} \exp(\text{sim}(\mathbf{h}_u^*, \mathbf{h}_p^*)/\tau)}, \quad (9)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, $\mathbf{h}_p^*$ denotes the embedding of the item belonging to another positive pair p in the batch $\mathcal{B}$ (which acts as a negative sample) and τ is a temperature hyperparameter. Unlike standard InfoNCE formulations that use augmented views of the same node as positives, the NA loss treats user–item interaction pairs as positive pairs: the numerator draws the user embedding closer to the embeddings of its interacted items, while the denominator pushes it away from all other user–item positive pairs in the batch. As shown by [16] through gradient analysis, this reformulation endows the contrastive objective with an intrinsic neighborhood aggregation capability equivalent to one step of graph convolution, while simultaneously preventing the over-smoothing that can arise from repeated message passing layers.

By combining BPR, NA loss, and L2 regularization in Eq. (7), the model jointly optimizes ranking accuracy (BPR), neighborhood-level representation quality (NA loss), and embedding generalization (regularization). The trainable parameters optimized during training are the context embedding matrix $\mathbf{E}_f$, the projection matrix $\mathbf{W}_c, \mathbf{b}_c$ of Eq. (2) and the base user/item embeddings.

4 Context-Sensitive Recommendation Evaluation Metrics

Standard ranking metrics like nDCG, Recall, and MAP measure whether a system retrieves items that have been interacted with by a user but do not take into account if these items fit the situational conditions of the recommendation request. We develop a set of of-line evaluation metrics that measure contextual appropriateness, which can be done using only the features from the dataset, without additional labeling or annotation.

We categorize the metrics into three groups, each focusing on a different aspect of quality of contextual recommendation that is time standard ranking methods are not able to see.

Family 1 – Context Similarity and Satisfaction Metrics measure the degree of set-based or geometric alignment between the query context c_q (the situational conditions under which recommendations are requested) and a representative context c_i of each recommended item. Five metrics are defined, each capturing a distinct aspect of this alignment: **The representative context is important. Should we mention something else about it?**

ACC@K (Average Context Consistency) measures the fraction of recommended items whose contextual attributes *exactly* match the

Table 1: Complementarity of the context similarity metrics.

Metric	Approach	Focus
ACC	Exact matching	Strict consistency
CS	Jaccard + penalty	Partial set overlap
WCS	Weighted Jaccard	Rare-feature alignment
WCA	IDF-weighted cosine	Geometric alignment
Friction	Complement of Hamming	Feature match rate

query context across all feature dimensions, with no partial credit awarded for near matches:

$$\text{ACC@K} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{K} \sum_{i \in R_q^K} \mathbf{1}_{\text{match}}(i, q) \quad (10)$$

where $\mathbf{1}_{\text{match}}(i, q)$ is an indicator function that equals 1 if item i exactly matches the context of query q across all feature dimensions, and 0 otherwise. **Here, i needs to be represented in terms of contextual features $f \in F$. That is the “representative context” of i .**

CS (Context Satisfaction) [22] relaxes the exactness requirement by evaluating partial feature overlap through a modified Jaccard coefficient:

$$\text{CS}(c_q, c_i, \alpha) = \frac{|c_q \cap c_i|}{|c_q \cup c_i| + \alpha \cdot \frac{|c_q \setminus c_i|}{|c_q|}} \quad (11)$$

where $\alpha \geq 0$ controls the penalty for query features absent from the item context. Setting $\alpha = 0$ recovers standard Jaccard similarity; in all experiments $\alpha = 0.5$ is used as a moderate penalty.

WCS (Weighted Context Satisfaction) extends CS by incorporating IDF-based feature importance weights w_f , so that rare, discriminative features contribute more than common ones:

$$\text{WCS}(c_q, c_i, w, \alpha) = \frac{\sum_f w_f \cdot \mathbf{1}_{f \in c_q \cap c_i}}{\sum_f w_f \cdot \mathbf{1}_{f \in c_q \cup c_i} + \alpha \cdot \frac{\sum_f w_f \cdot \mathbf{1}_{f \in c_q \setminus c_i}}{\sum_f w_f \cdot \mathbf{1}_{f \in c_q}}} \quad (12)$$

WCA (Weighted Context Alignment) measures the geometric alignment between query and item context vectors using IDF-weighted cosine similarity:

$$\text{WCA}(c_q, c_i, w) = \frac{\sum_f w_f \cdot c_{q,f} \cdot c_{i,f}}{\sqrt{\sum_f w_f \cdot c_{q,f}^2} \cdot \sqrt{\sum_f w_f \cdot c_{i,f}^2}} \quad (13)$$

WCA captures directional alignment in the feature embedding space, independently of feature cardinality.

Friction quantifies the proportion of contextual features that take the same value in both the query context and the item context, formulated as the complement of the normalized Hamming distance:

$$\text{Friction}(c_q, c_i) = 1 - \frac{d_H(c_q, c_i)}{F} = \frac{1}{F} \sum_{f=1}^F \mathbf{1}[c_{q,f} = c_{i,f}] \quad (14)$$

where F is the total number of contextual features and $d_H(c_q, c_i)$ is the Hamming distance between the two context vectors. A value of 1 indicates complete feature-level agreement; a value of 0 indicates total mismatch. Friction@K is then averaged over all top- K recommended items and all test queries:

$$\text{Friction@K} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{K} \sum_{i \in R_q^K} \text{Friction}(c_q, c_i) \quad (15)$$

Table 1 summarizes the five metrics and their complementary roles.

Family 2 – Advanced Context Dynamics analyze structural and distributional properties of the recommendation list with respect to contextual quality, beyond per-item alignment:

CR@K (Context Recall) shifts the evaluation focus from items to features. It measures the average fraction of query context features covered by each item in the top- K list:

$$\text{CR@K} = \frac{1}{K} \sum_{i \in R_K} \frac{|c_q \cap c_i|}{|c_q|} \quad (16)$$

$\text{CR@K} = 1$ indicates that every recommended item covers all requested features on average; unlike CS, it does not penalize extra features in the item context.

CGB (Context Group Balance) detects dimensional bias across predefined feature groups $\mathcal{G} = \{G_1, \dots, G_n\}$. For each group g , a group-specific context recall $\text{CR}_g@K$ is computed; CGB then measures how evenly these group recalls are distributed:

$$\text{CGB@K} = 1 - \min \left(\frac{\sigma_{\text{groups}}}{\sigma_{\text{max}}}, 1 \right) \quad (17)$$

where σ_{groups} is the standard deviation of group recalls and $\sigma_{\text{max}} = \sqrt{(|\mathcal{G}| - 1)/|\mathcal{G}|^2}$ is the theoretical maximum standard deviation, ensuring $\text{CGB@K} \in [0, 1]$. $\text{CGB} \approx 1$ indicates balanced contextual coverage across all groups; $\text{CGB} \approx 0$ indicates that the model excels at one dimension while neglecting the others.

Family 3 – Context-Weighted Ranking Metrics integrate contextual appropriateness directly into the relevance formulation of classical ranking metrics. Standard binary or graded relevance $\text{rel}(k)$ is replaced by context-weighted relevance:

$$\text{rel}_{\text{cw}}(k) = \text{rel}(k) \cdot C(c_q, c_{i_k}, \alpha) \quad (18)$$

where $C(\cdot)$ is any of the context similarity functions defined above (CS, WCA, or Friction). This ensures that each item contributes to the metric score in proportion to both user preference and contextual appropriateness.

CW-nDCG@K substitutes context-weighted relevance into the DCG gain function:

$$\text{CW-nDCG@K} = \frac{1}{\text{IDCG}_{\text{cw}}@K} \sum_{k=1}^K \frac{2^{\text{rel}_{\text{cw}}(k)} - 1}{\log_2(k+1)} \quad (19)$$

where $\text{IDCG}_{\text{cw}}@K$ is computed by sorting items jointly by relevance and context score in descending order. A score of 1 is achieved only when the system reproduces the ideal context-aware ranking. A large gap between nDCG and CW-nDCG indicates that the model retrieves relevant items but places them in positions where their contextual appropriateness is suboptimal.

CW-MAP@K incorporates context-weighted relevance into the average precision calculation:

$$\text{CW-MAP@K} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|\text{Rel}_q|} \sum_{k=1}^K \frac{\sum_{j=1}^k \text{rel}_{\text{cw}}(j)}{k} \cdot \text{rel}_{\text{cw}}(k) \quad (20)$$

A substantial decrease in CW-MAP relative to standard MAP signals that the model retrieves relevant items but does so in contextually inappropriate situations. Conversely, comparable CW-MAP and MAP scores indicate effective alignment between recommended items and the query context.

All metrics are bounded in $[0, 1]$, model-agnostic, and computable without annotations beyond the contextual features already present

in the dataset. Each metric is interpretable in isolation and, critically, captures a diagnostic dimension invisible to the others: together they provide a multi-faceted characterization of contextual recommendation quality that no single metric or subset can replicate.

5 Experiments

5.1 Datasets

We evaluate our approach on three benchmark datasets spanning distinct recommendation domains, context space structures, and interaction densities.

Frappe [23] is a mobile application recommendation dataset collected from a context-aware app discovery service. It comprises 96,203 implicit positive interactions between 957 users and 4,082 applications. Eight contextual features are organized into three semantic groups: *temporal* (daytime period, weekday, and a binary weekend indicator), *activity* (homework and data-cost concern), and *environment* (weather condition, country, and city). The combination of these features yields 5,382 unique context profiles. Because city and country alone account for thousands of distinct values, the feature space is highly sparse—a property that directly challenges context-aware models.

Yelp [24] is a large-scale local business recommendation dataset drawn from the Yelp Open Dataset. After preprocessing, it contains 604,498 interactions (reviews with ≥ 4 stars treated as positive) across 45,651 users and 16,237 businesses. Eleven contextual features are organized into three groups: *temporal* (hour of day, day of week, weekend indicator), *social and user* (review length, elite status, user experience level, alcohol availability, and outdoor seating), and *spatial* (city with 637 unique values, business category with 340 unique values, and price range). The high cardinality of the spatial features produces approximately 150 K unique context profiles, making Yelp the most contextually complex dataset in our study.

BoardGameGeek (BGG) [25] captures user ratings for board games enriched with explicit contextual annotations collected via user self-reporting on the BoardGameGeek community platform. After applying a relevance threshold of rating ≥ 7 , the dataset contains 1,113,609 interactions across 43,660 users and 901 items. Twenty-one binary contextual features are organized into three semantic groups: *Playing Time* (very short, short, medium, long, very long), *Gaming Mood* (party, easy-going, expert, intense, cooperative, competitive, thematic, story-based), and *Social Companion* (solo, two players, large group, toddlers, preschoolers, children, family, friends). Because features are binary and the catalog is small, the space of distinct context profiles is compact at 410 unique contexts. BGG serves as an ideal testbed for context-aware evaluation: the gaming situation fundamentally determines which game is appropriate, providing a clear ground truth for contextual relevance.

5.2 Baseline Recommenders

We compare LightCCF-CA against three non-contextual baselines and five state-of-the-art context-aware FM-based models, all trained within the WarpRec framework [26].

Non-contextual baselines. *Random* assigns uniform random scores to all items independently of user identity and context, representing the theoretical lower bound. *Popularity* scores items by their global

interaction count in the training split and recommends the most popular items to every user; despite its simplicity, it is a strong baseline in domains where a small item set dominates aggregate preferences. The third non-contextual baseline is *LightCCF*, the graph-based recommender that we modified to derive our model.

Context-aware FM-based models. Following the contextual modeling paradigm discussed in Sec. 2.1, we select five representative FM-based models as context-aware baselines. *FM* [9] models all pairwise feature interactions through shared latent factors, enabling generalization to unobserved feature co-occurrences. *DeepFM* [10] combines an FM component for second-order interactions with a deep MLP for higher-order interactions, sharing embeddings between both branches. *NFM* [11] replaces FM’s inner-product pooling with a Bi-Interaction pooling layer followed by an MLP, enabling non-linear modeling of pairwise feature interactions. *AFM* [12] extends FM with a neural attention network that learns a scalar importance weight for each pairwise feature interaction, allowing the model to suppress uninformative feature pairs. *xDeepFM* [13] introduces a Compressed Interaction Network (CIN) that models bounded-degree feature interactions at the vector level, combined with a standard MLP component.

5.3 Evaluation Methodology

All models are evaluated under a *full-catalog ranking* strategy [18]: at inference time every item in the catalog is scored for each test query, eliminating sampling bias. Items already observed during training are masked with a score of $-\infty$ before ranking, so that evaluation measures the model’s ability to discover new relevant items.

Data splitting. We adopt a temporal leave-one-out protocol: the most recent interaction of each user forms the test set, the second-most-recent forms the validation set, and all earlier interactions constitute the training set. This simulates a realistic online recommendation scenario and prevents any leakage of future data [27–29].

Context in evaluation. As shown in Eq. 6, the score is computed including the product between the context embedding and the item embedding. During evaluation, the contextual features associated with the held-out interaction (i.e., the context associated to the interaction between user and positive item in the test set) are provided as input to the context encoder of the model, in order to obtain the context embedding used to compute the score for both positive (the held-out item) and negative items (all others items in the catalog). This setup simulates a realistic deployment scenario in which the user’s current context (e.g., time of day, location) is known at query time [30]

Hyperparameter optimization. Hyperparameters are selected via Ray Tune [31] using distributed search over the validation set. Training is governed by early stopping that monitors nDCG@10 with a patience of 10 epochs and a grace period of 20 epochs. The best checkpoint is retrained on the full training split before final test evaluation. All models use an embedding size of 64 (reduced to 16 for xDeepFM due to its quadratic CIN memory cost and reduced to 32 for LightCCF-CA on BGG dataset), a batch size of 2048 (512 for xDeepFM), a learning rate of 10^{-3} , a regularization weight of 10^{-3} , and 2 negative samples per positive interaction.

Evaluation cutoffs. All metrics are evaluated at cutoffs $K \in \{5, 10\}$, reflecting typical practical recommendation-list lengths. In Sec. 5.4 we report results at cutoff $K = 10$.

5.4 Results

As stated in Sec. 5.2, we evaluate our model against a set of baseline recommenders comprising both standard collaborative filtering methods and context-aware recommendation models. For context-aware baselines, we report results obtained using the full set of available contextual features for each dataset, as provided by the respective dataset sources. For our proposed model, we report results obtained using the subset of contextual features that yields the highest nDCG@10 on the validation set; the selection of this optimal subset is described in detail in the ablation study (Sec. 5.5). Results are reported in terms of both standard ranking metrics (nDCG, Recall, Precision, MRR) and context-aware metrics.

5.4.1 Standard Ranking Metrics. Tables 2–4 report nDCG@10, Recall@10, Precision@10, and MRR@10 for all models on the Yelp, Frappe, and BoardGameGeek datasets respectively.

Table 2: Traditional Ranking Metrics on Yelp. Bold values indicate the best-performing model.

Model	nDCG	MRR	Precision	Recall
Random	0.0004	0.0002	0.0001	0.0009
Pop	0.0162	0.0110	0.0033	0.0337
LightCCF	0.0510	0.0365	0.0099	0.0991
AFM	0.0418	0.0303	0.0080	0.0799
DeepFM	0.0151	0.0100	0.0032	0.0321
FM	0.0395	0.0279	0.0078	0.0780
NFM	0.0225	0.0160	0.0044	0.0441
xDeepFM	0.0226	0.0152	0.0047	0.0473
LightCCF-CA	0.0540	0.0393	0.0103	0.1029

On the **Yelp** dataset, our model achieves the best performance across nDCG@10, MRR@10, Precision@10 and Recall@10, outperforming the strongest baseline, LightCCF, by +5.9% in nDCG@10, +7.7% in MRR@10, +4.0% in Precision@10, and +3.8% in Recall@10. Among the context-aware baselines, AFM ranks second in nDCG@10 and Recall@10, while factorization-based models (FM, AFM) consistently outperform deep architectures (DeepFM, NFM, xDeepFM) on this dataset. Notably, LightCCF (the non-contextual graph-based baseline from which our model is derived) already represents the strongest competitor on Yelp, suggesting that the graph-based collaborative signal is particularly informative for venue recommendation. The improvement brought by our context-aware extension confirms that integrating contextual features into both the message-passing phase and the score computation yields a measurable gain over the non-contextual graph baseline.

Table 3: Traditional Ranking Metrics on Frappe.

Model	nDCG	MRR	Precision	Recall
Random	0.0000	0.0000	0.0001	0.0000
Pop	0.0420	0.0318	0.0077	0.0749
LightCCF	0.0397	0.0302	0.0071	0.0711
AFM	0.0427	0.0324	0.0099	0.0762
DeepFM	0.0405	0.0299	0.0078	0.0685
FM	0.0509	0.0379	0.0090	0.0930
NFM 0.2549	0.0312	0.0224	0.0069	0.0607
xDeepFM	0.0310	0.0247	0.0076	0.0517
LightCCF-CA	0.0543	0.0404	0.0099	0.0995

On the **Frappe** dataset, our model again achieves the highest nDCG@10, MRR@10, and Recall@10, with improvements of +6.7%, +6.6%, and +7.0% over FM, the strongest non-graph baseline. Precision@10 is tied with AFM at 0.0099. A notable difference with respect to Yelp is the behavior of LightCCF: on Frappe, the graph-based baseline underperforms compared to factorization models (FM achieves nDCG@10 = 0.0509 against LightCCF's 0.0397), suggesting that the collaborative graph structure alone is less effective on this dataset, possibly due to the sparser and more heterogeneous nature of mobile application usage compared to venue check-ins. LightCCF-CA substantially closes this gap and surpasses FM, indicating that contextual features compensate for the weaker collaborative signal in the graph.

Table 4: Traditional Ranking Metrics on BGG.

Model	nDCG	MRR	Precision	Recall
Random	0.0052	0.0033	0.0011	0.0114
Pop	0.0972	0.0708	0.0186	0.1861
LightCCF	0.1259	0.0914	0.0240	0.2401
AFM	0.1245	0.0901	0.0238	0.2382
DeepFM	0.1170	0.0841	0.0226	0.2257
FM	0.1146	0.0818	0.0223	0.2230
NFM	0.1164	0.0835	0.0225	0.2251
xDeepFM	0.1087	0.0789	0.0207	0.2072
LightCCF-CA	0.1218	0.0878	0.0234	0.2341

On the **BGG** dataset, our model ranks third overall, behind LightCCF and AFM, with a gap of −3.3% in nDCG@10 and −2.5% in Recall@10 with respect to the best-performing model, LightCCF. It is important to note that in this comparison, all baselines were trained with an embedding size of 64 (except xDeepFM, as we said in Sec. 5.3), while LightCCF-CA was constrained to an embedding size of 32 due to GPU memory limitations caused by the size of the BGG dataset. The high number of binary contextual features, combined with the context-aware message passing on every edge of the graph, results in a significantly higher memory footprint compared to the other two datasets, making it impossible to match the embedding capacity of the baselines.

Despite this limitation, LightCCF-CA outperforms all factorization-based baselines (DeepFM, FM, NFM, xDeepFM) and ranks competitively above six out of eight baselines. The performance gap with respect to LightCCF (trained at double the embedding size) is limited to 3.3% in nDCG@10, which suggests that the contextual augmentation partially compensates for the reduced representational capacity.

5.4.2 Context-Aware Metrics. In addition to the standard ranking evaluation, this section reports context-aware metrics (introduced in Sec. 4) that measure how well each system adapts its recommendations to the situational conditions of each query. Tables 5 and 7 report these metrics for all models in the three datasets.

Across all three datasets, LightCCF-CA gets better or comparable with other competitors on most of the context-aware metrics, which indicates that by incorporating contextual information into the graph propagation mechanism, we can generate more accurate recommendations for specific query conditions.

On **Yelp**, LightCCF-CA ranks first on CS@10, WCS@10, WCA@10, Friction@10, CR@10, CW-nDCG@10, and CW-MAP@10. This confirms that its recommendations rank relevant items higher and do so in a way that is more fitting for the context. The only metric where

it does not lead is CGB@10, where DeepFM achieves the highest score (0.9328). As discussed in Sec. 4, CGB measures the balance of contextual coverage across predefined feature groups; a model that distributes recommendations broadly across groups without specializing in the query context can achieve high CGB by accident rather than by design. This interpretation is consistent with DeepFM's substantially lower performance on all other context-aware metrics. On **Frappe**, LightCCF-CA obtains the best results on CS@10, WCS@10, Friction@10, CR@10, CGB@10, CW-nDCG@10, and CW-MAP@10 with the strongest lead on all the three datasets. The difference with FM, the best FM-based baseline, is small but in favour of LightCCF-CA for all the cases, indicating that context-gated message passing is able to capture contextual signals that pairwise feature interaction alone fails to reveal, even in a sparse dataset. The WCS analysis (Table 7) shows that LightCCF-CA achieves the best concordance on the Environment group (0.4346), which contains the highest-cardinality features (city, country) and thus offers the most informative contextual signal. ACC@10 is marginally below AFM and DeepFM (0.0023 vs. 0.0026), a direct consequence of all models on this dataset for which exact full-context matches are still quite infrequent.

On **BGG**, the results show a more complex picture. LightCCF-CA achieves the highest WCA@10 (0.3827) and CW-nDCG@10 (0.0971) results, which suggest that its recommendations are much better fitted to the query context geometrically and also contextually appropriate items are placed at higher rank positions. However, in the rest of the metrics (CS@10, WCS@10, Friction@10, CR@10, and CGB@10) Pop still outperforms LightCCF-CA. This result is, to some extent, a consequence of the smaller embedding size (32 vs. 64) which was enforced by GPU memory limitations, and thus implying representational capacity restriction relative to the baselines. In addition to this technical limitation, the high CGB values across all models (> 0.88) not only mirror a characteristic of the BGG context space but also highlight that the three feature groups (Playtime, Mood, Social) have each similar cardinality and since they are frequently interacted during training, models cover them with similar levels of recall, which makes CGB a less distinguishing signal on this dataset. The WCS analysis further corroborates that LightCCF-CA is best on the Playtime (0.7986) and Social (0.8554) groups, while Pop scores highest on Mood group (0.8078), which is in line with its preference for recommending popular items that perfectly match different gaming moods. LightCCF-CA, despite the embedding constraint, manages to beat all FM-based baselines on WCA@10 and CW-nDCG@10 and almost matches AFM on CW-MAP@10 with a difference of only 0.001 (0.0635 vs. 0.0645).

Overall. The context-aware evaluation complements the picture emerging from standard ranking metrics: on Yelp and Frappe, where LightCCF-CA already leads on accuracy, the proposed metrics confirm stronger contextual appropriateness as well. On BGG, where the embedding constraint limits standard ranking performance, context-aware metrics reveal that LightCCF-CA still achieves superior geometric alignment and contextual ranking quality, a dimension that accuracy measures alone would overlook.

5.5 Contextual Feature Ablation Study

To assess the contribution of different types of contextual information to recommendation quality of our proposed model, we conduct a

Table 5: Context-aware metrics at cutoff $K = 10$ on Yelp, Frappe, and BGG. Bold values indicate the best result. Fric. = Friction. WCA is excluded from Frappe (structurally saturated at 1.0).

Dataset	Model	Context Similarity & Satisfaction					Advanced Dynamics & Weighted Ranking			
		ACC	CS	WCS	WCA	Fric.	CR	CGB	CW-nDCG	CW-MAP
Yelp	Random	0.0000	0.3403	0.3985	0.8159	0.3403	0.3403	0.9201	0.0002	0.0001
	Pop	0.0000	0.3613	0.4113	0.8553	0.3613	0.3613	0.9257	0.0070	0.0031
	LightCCF	0.0000	0.3874	0.4180	0.8353	0.3874	0.3874	0.8920	0.0216	0.0100
	AFM	0.0000	0.3836	0.4169	0.8414	0.3836	0.3836	0.9164	0.0180	0.0084
	DeepFM	0.0000	0.3609	0.4127	0.8406	0.3609	0.3609	0.9328	0.0065	0.0027
	FM	0.0000	0.3851	0.4166	0.8437	0.3851	0.3851	0.9071	0.0169	0.0078
	NFM	0.0000	0.3692	0.4097	0.8317	0.3692	0.3692	0.9194	0.0093	0.0041
	xDeepFM	0.0000	0.3792	0.4171	0.8334	0.3792	0.3792	0.9091	0.0096	0.0041
	LightCCF-CA	0.0000	0.3879	0.4194	0.8557	0.3879	0.3879	0.9015	0.0228	0.0107
Frappe	Random	0.0014	0.5308	0.5669	—	0.5308	0.5308	0.4983	0.0000	0.0000
	Pop	0.0021	0.5401	0.5778	—	0.5401	0.5401	0.5063	0.0196	0.0100
	LightCCF	0.0023	0.5383	0.5730	—	0.5383	0.5383	0.5150	0.0187	0.0095
	AFM	0.0026	0.5404	0.5773	—	0.5404	0.5404	0.5095	0.0206	0.0108
	DeepFM	0.0026	0.5391	0.5765	—	0.5391	0.5391	0.5042	0.0192	0.0103
	FM	0.0025	0.5405	0.5772	—	0.5405	0.5405	0.5133	0.0251	0.0132
	NFM	0.0012	0.5317	0.5688	—	0.5317	0.5317	0.5119	0.0151	0.0076
	xDeepFM	0.0013	0.5316	0.5690	—	0.5316	0.5316	0.5089	0.0145	0.0075
	LightCCF-CA	0.0023	0.5422	0.5782	—	0.5422	0.5422	0.5166	0.0262	0.0134
BGG	Random	0.0012	0.7758	0.7758	0.2762	0.7758	0.7758	0.8882	0.0033	0.0022
	Pop	0.0009	0.8159	0.8159	0.3576	0.8159	0.8159	0.9445	0.0708	0.0502
	LightCCF	0.0018	0.8067	0.8067	0.3818	0.8067	0.8067	0.9211	0.0965	0.0625
	AFM	0.0017	0.8074	0.8074	0.3737	0.8074	0.8074	0.9247	0.0901	0.0645
	DeepFM	0.0018	0.8052	0.8052	0.3738	0.8052	0.8052	0.9202	0.0841	0.0601
	FM	0.0019	0.8051	0.8051	0.3692	0.8051	0.8051	0.9218	0.0818	0.0584
	NFM	0.0013	0.8089	0.8089	0.3612	0.8089	0.8089	0.9295	0.0835	0.0599
	xDeepFM	0.0016	0.8034	0.8034	0.3655	0.8034	0.8034	0.9206	0.0789	0.0564
	LightCCF-CA	0.0018	0.8095	0.8095	0.3827	0.8095	0.8095	0.9231	0.0971	0.0635

WCA is excluded from Frappe (all models saturate at 1.0 due to binary one-hot encoding).

systematic ablation study across the three datasets employed in this work: Yelp, Frappe, and BoardGameGeek (BGG). For each dataset, the available contextual features are organized into semantically coherent groups, and multiple feature subsets are evaluated independently to isolate the effect of each group on nDCG@10.

5.5.1 Yelp. The Yelp dataset provides twelve contextual features, which we partition into three semantic groups. *Temporal* features encode when an interaction occurs; *Location* features describe venue-specific properties; *Atmosphere* features capture environmental attributes of the venue. The features `hour_of_day`, `user_elite`, and `user_experience` are excluded from all subsets: `hour_of_day` is subsumed by `time_slot`, while `user_elite` and `user_experience` represent static user attributes rather than situational context. Given these considerations, the three main subsets are:

- **Temporal** (4 features): `day_of_week`, `is_weekend`, `season`, `time_slot`
- **Location** (3 features): `city`, `category`, `price_range`
- **Atmosphere** (2 features): `alcohol`, `outdoor_seating`

Eight configurations are evaluated, corresponding to individual subsets and their pairwise combinations, plus a complete set (ALL) that includes all twelve features. The configurations and their relative ranking by nDCG@10 are summarized in Table 9. The combination of Temporal and Location features (SUB6) achieves the highest NDCG@10, followed by Location alone (SUB3) and the near-complete set SUB1, which includes all groups but excludes the user-centric attributes. The full feature set (ALL) yields the lowest performance among all evaluated configurations, suggesting

that the inclusion of redundant or noisy features can impair the model’s ability to leverage contextual signals effectively. This finding aligns with the intuition that, for business recommendation, *where* and *when* a visit occurs are the most informative contextual dimensions, confirming the results of prior works ([32], [33] and [34]) where spatial and temporal context have been shown to be the most informative signals among all features of Yelp dataset.

5.5.2 Frappe. The Frappe dataset includes eight contextual features describing the situational context of mobile application usage, which we can partition into three semantic groups. *Temporal* features encode when the mobile app is used; *Activity* features specify the conditions under which the mobile app is used; *Environment* features capture environmental attributes that occur during app usage. We organize them into three groups:

- **Temporal** (3 features): `daytime`, `weekday`, `isweekend`
- **Activity** (2 features): `homework`, `cost`
- **Environment** (3 features): `weather`, `country`, `city`

Seven configurations are evaluated, including the three individual groups, their pairwise combinations, and the complete feature set (ALL). Results are reported in Table 10. Environment features alone (SUB3) achieve the best nDCG@10, with the full feature set (ALL) ranking second, suggesting that environmental context is the dominant signal in this dataset, as partially demonstrated in literature [35]. The combination of Temporal and Activity features (SUB4) is the weakest configuration, indicating potential interference between these two groups when environmental grounding is absent.

Table 6: Same as Table 5 but in 1col. Context-aware metrics at cutoff $K = 10$ on Yelp, Frappe, and BGG. Bold values indicate the best result. Fric. = Friction. WCA is excluded from Frappe (structurally saturated at 1.0).

(a) Yelp									
Model	Context Similarity & Satisfaction					Advanced Dynamics & Weighted Ranking			
	ACC	CS	WCS	WCA	Fric.	CR	CGB	CW-nDCG	CW-MAP
Random	0.0000	0.3403	0.3985	0.8159	0.3403	0.3403	0.9201	0.0002	0.0001
Pop	0.0000	0.3613	0.4113	0.8553	0.3613	0.3613	0.9257	0.0070	0.0031
LightCCF	0.0000	0.3874	0.4180	0.8353	0.3874	0.3874	0.8920	0.0216	0.0100
AFM	0.0000	0.3836	0.4169	0.8414	0.3836	0.3836	0.9164	0.0180	0.0084
DeepFM	0.0000	0.3609	0.4127	0.8406	0.3609	0.3609	0.9328	0.0065	0.0027
FM	0.0000	0.3851	0.4166	0.8437	0.3851	0.3851	0.9071	0.0169	0.0078
NFM	0.0000	0.3692	0.4097	0.8317	0.3692	0.3692	0.9194	0.0093	0.0041
xDeepFM	0.0000	0.3792	0.4171	0.8334	0.3792	0.3792	0.9091	0.0096	0.0041
LightCCF-CA	0.0000	0.3879	0.4194	0.8557	0.3879	0.3879	0.9015	0.0228	0.0107

(b) Frappe									
Model	Context Similarity & Satisfaction					Advanced Dynamics & Weighted Ranking			
	ACC	CS	WCS	WCA	Fric.	CR	CGB	CW-nDCG	CW-MAP
Random	0.0014	0.5308	0.5669	—	0.5308	0.5308	0.4983	0.0000	0.0000
Pop	0.0021	0.5401	0.5778	—	0.5401	0.5401	0.5063	0.0196	0.0100
LightCCF	0.0023	0.5383	0.5730	—	0.5383	0.5383	0.5150	0.0187	0.0095
AFM	0.0026	0.5404	0.5773	—	0.5404	0.5404	0.5095	0.0206	0.0108
DeepFM	0.0026	0.5391	0.5765	—	0.5391	0.5391	0.5042	0.0192	0.0103
FM	0.0025	0.5405	0.5772	—	0.5405	0.5405	0.5133	0.0251	0.0132
NFM	0.0012	0.5317	0.5688	—	0.5317	0.5317	0.5119	0.0151	0.0076
xDeepFM	0.0013	0.5316	0.5690	—	0.5316	0.5316	0.5089	0.0145	0.0075
LightCCF-CA	0.0023	0.5422	0.5782	—	0.5422	0.5422	0.5166	0.0262	0.0134

(c) BGG									
Model	Context Similarity & Satisfaction					Advanced Dynamics & Weighted Ranking			
	ACC	CS	WCS	WCA	Fric.	CR	CGB	CW-nDCG	CW-MAP
Random	0.0012	0.7758	0.7758	0.2762	0.7758	0.7758	0.8882	0.0033	0.0022
Pop	0.0009	0.8159	0.8159	0.3576	0.8159	0.8159	0.9445	0.0708	0.0502
LightCCF	0.0018	0.8067	0.8067	0.3818	0.8067	0.8067	0.9211	0.0965	0.0625
AFM	0.0017	0.8074	0.8074	0.3737	0.8074	0.8074	0.9247	0.0901	0.0645
DeepFM	0.0018	0.8052	0.8052	0.3738	0.8052	0.8052	0.9202	0.0841	0.0601
FM	0.0019	0.8051	0.8051	0.3692	0.8051	0.8051	0.9218	0.0818	0.0584
NFM	0.0013	0.8089	0.8089	0.3612	0.8089	0.8089	0.9295	0.0835	0.0599
xDeepFM	0.0016	0.8034	0.8034	0.3655	0.8034	0.8034	0.9206	0.0789	0.0564
LightCCF-CA	0.0018	0.8095	0.8095	0.3827	0.8095	0.8095	0.9231	0.0971	0.0635

5.5.3 *BoardGameGeek (BGG)*. The BGG dataset is enriched with 21 binary contextual features, grouped into three high-level categories that characterize the situational context of a board game session:

- **Playtime** (5 features): very_short, short, moderate, long, very_long
- **Mood** (8 features): party, easy-going, expert, intense, cooperative, competitive, thematic, story-based
- **Social** (8 features): 1-player, 2-players, large-group, toddlers, preschoolers, children, family, friends

Seven configurations are evaluated: the three individual groups, their pairwise combinations, and the complete feature set (ALL). Results are reported in Table 11. The combination of Playtime and Mood (SUB4) achieves the highest nDCG@10, followed by Social alone (SUB3) and Playtime combined with Social (SUB5). Playtime alone (SUB1) yields the weakest performance, suggesting that session duration is most informative only when paired with additional situational context such as gaming mood.

It is worth noting that, for context-aware baselines described in Sec. 5.2, performance on standard ranking metrics does not vary across different subsets of contextual features. We hypothesize that this is due to the architectural design of such models, in which context influences either the score computation or the training objective, but not the propagation of information through the interaction graph. In contrast, LightCCF-CA integrates contextual features into both

the message passing phase (Eq. 3) and the score computation (Eq. 6), making the propagated embeddings themselves context-dependent and thus more sensitive to the choice of contextual features.

5.6 Qualitative

Analysis: Context-Aware Validation Signal

Beyond the quantitative evaluation reported in the previous section, we conduct a qualitative analysis to examine whether optimizing the validation objective with CW-nDCG@10 influences the ranking behavior of LightCCF-CA at the individual user level. Specifically, we compare two training configurations for each dataset: (i) the model validated on standard nDCG@10 using the best performing context feature subset identified in Section 5.5 (hereafter BASE), and (ii) the model validated on CW-nDCG@10 under identical hyperparameters and feature configuration (CA-VAL). The top-10 recommendation lists produced by BASE and CA-VAL are compared against the ground truth, so that a user is considered a *hit* if their positive item appears among the ten recommended items. Figure 1 reports the hit rates the three datasets.

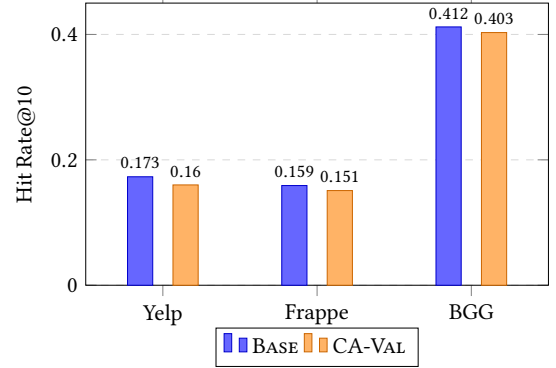


Figure 1: Comparison of Hit Rate@10 under standard (BASE) and context-aware (CA-VAL) validation across three datasets.

Across all three datasets, CA-VAL yields a modest net reduction in Hit Rate@10 relative to BASE (Yelp: -1.3% , Frappe: -0.8% , BGG: -0.9%).

Context-Aware Metric Analysis. To further investigate whether the context-aware validation signal influences the contextual quality of recommendations, Table 12 compares the two configurations across five context-similarity metrics.

On Yelp and Frappe, CA-VAL achieves marginal improvements in CS@10 and WCS@10, indicating that the model trained with context-aware validation produces recommendations whose contextual attributes align slightly more closely with the user’s query context. Specifically, on Yelp, CA-VAL reaches CS@10 = 0.3916 (+0.95% over BASE) and WCS@10 = 0.4228 (+0.81%), while on Frappe the improvements are CS@10 = 0.5429 (+0.13%) and WCS@10 = 0.5790 (+0.14%). However, on BGG, the trend reverses: BASE outperforms CA-VAL on all metrics (CS@10 = 0.8095 vs. 0.8029, WCS@10 = 0.8095 vs. 0.8029, WCA@10 = 0.3827 vs. 0.3798). Taken together, the hit rate degradation (Figure 1) and the mixed contextual metric performance (Table 12) suggest that the context-aware validation signal provides limited practical benefit. While CA-VAL achieves modest gains in

Table 7: Weighted Context Satisfaction (WCS) at cutoff $K=10$ breakdown per context group.

Model	BGG			Frappe			Yelp		
	Mood	Playtime	Social	Activity	Envir.	Temp.	Business	Social	Temp.
Random	0.7086	0.7854	0.8369	0.9070	0.4190	0.4296	0.4514	0.4074	0.3410
Pop	0.8078	0.7817	0.8454	0.9101	0.4231	0.4518	0.4527	0.4224	0.3639
LightCCF	0.7644	0.7981	0.8545	0.8974	0.4325	0.4386	0.4790	0.4103	0.3495
AFM	0.7685	0.7958	0.8536	0.9082	0.4295	0.4469	0.4704	0.4202	0.3619
DeepFM	0.7625	0.7961	0.8536	0.9109	0.4264	0.4451	0.4684	0.4107	0.3584
FM	0.7637	0.7953	0.8527	0.9055	0.4320	0.4467	0.4744	0.4207	0.3567
NFM	0.7766	0.7901	0.8528	0.8976	0.4208	0.4400	0.4650	0.4145	0.3520
xDeepFM	0.7612	0.7934	0.8517	0.8997	0.4169	0.4419	0.4867	0.4116	0.3508
LightCCF-CA	0.7679	0.7986	0.8554	0.9047	0.4346	0.4476	0.4851	0.4182	0.3547

Table 8: Same as Table 7 but in 1col – another option is to show 3 subtables, one per dataset (or two of them can be presented together...) Weighted Context Satisfaction (WCS) at cutoff $K=10$ breakdown per context group.

Model	BGG			Frappe			Yelp		
	Mood	Playtime	Social	Activity	Envir.	Temp.	Business	Social	Temp.
Random	0.7086	0.7854	0.8369	0.9070	0.4190	0.4296	0.4514	0.4074	0.3410
Pop	0.8078	0.7817	0.8454	0.9101	0.4231	0.4518	0.4527	0.4224	0.3639
LightCCF	0.7644	0.7981	0.8545	0.8974	0.4325	0.4386	0.4790	0.4103	0.3495
AFM	0.7685	0.7958	0.8536	0.9082	0.4295	0.4469	0.4704	0.4202	0.3619
DeepFM	0.7625	0.7961	0.8536	0.9109	0.4264	0.4451	0.4684	0.4107	0.3584
FM	0.7637	0.7953	0.8527	0.9055	0.4320	0.4467	0.4744	0.4207	0.3567
NFM	0.7766	0.7901	0.8528	0.8976	0.4208	0.4400	0.4650	0.4145	0.3520
xDeepFM	0.7612	0.7934	0.8517	0.8997	0.4169	0.4419	0.4867	0.4116	0.3508
LightCCF-CA	0.7679	0.7986	0.8554	0.9047	0.4346	0.4476	0.4851	0.4182	0.3547

Table 9: Yelp contextual feature subsets, ordered by nDCG@10 (descending).

Rank	ID	Feature Groups	nDCG	CW-nDCG
1	SUB6	Temporal + Location	0.0540	0.0228
2	SUB3	Location	0.0512	0.0217
3	SUB1	Temporal + Location + Atmosphere	0.0493	0.0209
4	SUB5	Location + Atmosphere	0.0469	0.0199
5	SUB7	Temporal + Atmosphere	0.0415	0.0176
6	SUB4	Atmosphere	0.0409	0.0172
7	SUB2	Temporal	0.0365	0.0154
8	ALL	All 12 features	0.0329	0.0138

Table 10: Frappe contextual feature subsets.

Rank	ID	Feature Groups	nDCG	CW-nDCG
1	SUB3	Environment	0.0543	0.0262
2	ALL	Environment + Temporal + Activity	0.0523	0.0255
3	SUB6	Activity + Environment	0.0511	0.0254
4	SUB2	Activity	0.0492	0.0240
5	SUB5	Temporal + Environment	0.0483	0.0235
6	SUB1	Temporal	0.0481	0.0235
7	SUB4	Temporal + Activity	0.0476	0.0235

contextual alignment on Yelp and Frappe, these improvements do not translate into better retrieval coverage or ranking quality, and the signal degrades performance on BGG.

User-Level Ranking Behavior. To further examine how the validation objective affects individual ranking decisions, the two hit sets are joined on user_id, yielding three disjoint user subsets:

- $\mathcal{U}_{\text{both}}$: users whose positive item appears in the top-10 under both configurations;
- $\mathcal{U}_{\text{base}}$: users whose positive item appears in the top-10 only under BASE;

Table 11: BGG contextual feature subsets.

Rank	ID	Feature Groups	nDCG	CW-nDCG
1	SUB4	Playtime + Mood	0.1218	0.0971
2	SUB3	Social	0.1217	0.0966
3	SUB5	Playtime + Social	0.1213	0.0960
4	ALL	All 21 features	0.1197	0.0949
5	SUB2	Mood	0.1178	0.0947
6	SUB6	Mood + Social	0.1019	0.0805
7	SUB1	Playtime	0.0832	0.0653

Table 12: Comparison of context similarity & satisfaction metrics under standard (BASE) and context-aware (CA-VAL) validation. Fric. = Friction. WCA excluded from Frappe.

Dataset	Configuration	ACC	CS	WCS	WCA	Fric.
Yelp	BASE	0.0000	0.3879	0.4194	0.8557	0.3879
	CA-VAL	0.0000	0.3916	0.4228	0.8478	0.3916
Frappe	BASE	0.0023	0.5422	0.5782	—	0.5422
	CA-VAL	0.0026	0.5429	0.5790	—	0.5429
BGG	BASE	0.0018	0.8095	0.8095	0.3827	0.8095
	CA-VAL	0.0016	0.8029	0.8029	0.3798	0.8029

WCA is excluded from Frappe (all models saturate at 1.0 due to binary one-hot encoding).

- $\mathcal{U}_{\text{ca}}$: users whose positive item appears in the top-10 only under CA-VAL.

Users in $\mathcal{U}_{\text{ca}}$ represent cases where context-aware validation manages to retrieve a positive item that is not surfaced by the standard validation; users in $\mathcal{U}_{\text{base}}$ represent the symmetric case. For users in $\mathcal{U}_{\text{both}}$, we compare the exact rank of the positive item under the two configurations to assess whether CA-VAL promotes it to a higher position within the list. Note that this analysis operates at cutoff $K=10$ and a continuous rank level comparison is only possible for $\mathcal{U}_{\text{both}}$. Table 13 reports the rank based analysis on $\mathcal{U}_{\text{both}}$.

Table 13: Rank comparison on $\mathcal{U}_{\text{both}}$: fraction of users for whom CA-VAL promotes, demotes, or preserves the rank of the positive item relative to BASE.

Dataset	$ \mathcal{U}_{\text{both}} $	CA better	Base better	Equal
Yelp	1,589	556 (35.0%)	599 (37.7%)	434 (27.3%)
Frappe	171	60 (35.1%)	58 (33.9%)	53 (31.0%)
BGG	24,127	9,237 (38.3%)	9,854 (40.8%)	5,036 (20.9%)

This analysis on $\mathcal{U}_{\text{both}}$ reveals a broadly symmetric behavior: on Yelp and BGG, BASE promotes the positive item to a higher position slightly more often than CA-VAL (37.7% vs. 35.0% on Yelp; 40.8% vs. 38.3% on BGG), while on Frappe the two configurations are nearly

indistinguishable (33.9% vs. 35.1%). A non negligible fraction of users (27.3% on Yelp, 31.0% on Frappe, and 20.9% on BGG) exhibit identical ranks under both configurations, suggesting that for a substantial portion of users the validation objective has no observable effect on the final ranking.

These results indicate that using CW-nDCG@10 as the validation metric does not improve recommendation quality as measured by either hit rate or rank position of the positive item. The slight degradation in both metrics, coupled with the limited improvement in context-aware metrics, suggests that the context-aware validation signal, as currently formulated, does not provide a learning advantage over the standard nDCG@10. One possible explanation is rooted in the fact that context-aware methodologies inherently rely on typical usage patterns to assess relevance [2]. Consequently, enforcing a context-aware metric like CW-nDCG@10 during validation may overly penalize contextually misaligned user-item pairs. This strict penalty can lead the model to down rank items that are genuinely relevant to the user but happen to occur in atypical contexts within the test set. For example, if a user visits a restaurant at an unusual time or in an unexpected location, the context-aware validation signal would assign a lower reward to retrieving that item, even though it represents a ground-truth positive interaction.

6 Conclusions

This work introduced LightCCF-CA, a context-aware GNN-based recommendation model that integrates contextual information directly into graph propagation via context-gated message passing, and a comprehensive suite of nine context-sensitive evaluation metrics that capture aspects of recommendation quality invisible to traditional accuracy measures. Experimental results on three benchmark datasets demonstrate that LightCCF-CA achieves competitive or superior performance on standard ranking metrics while maintaining stronger contextual alignment as measured by the proposed metrics. The contextual feature ablation study reveals that the optimal subset of context features is domain-dependent: temporal and spatial features dominate on Yelp, environmental features on Frappe, and playtime combined with mood on BGG. Our qualitative analysis of context-aware validation signals indicates that using CW-nDCG@10 as the validation objective does not yield measurable improvements over standard nDCG@10, suggesting that further refinements to the context-weighted ranking formulation may be necessary.

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